Python Fundamentals (Assignment1)

Assignment Problems

Q1. Write a program that asks the user for their name and age, then prints a sentence like:

```
Hello Shradha, you are 21 years old!
```

Q2. Take two numbers as input from the user and print their **sum, difference, product, and quotient**.

Q3. Ask the user to enter two integers and one float. Convert them all to floats and print their average.

Q4. The user enters a string containing a number (e.g., `"45"`). Convert it to:

- an integer
- a float
- a string again

  Print all three values with their types.

Q5. Evaluate and print the result of the following expression:

```
x = 10 + 3 * 2 ** 2
```

Based on what you learnt in the lecture explain why the output is what it is.

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Q6. Write a program to **swap** values of two numbers entered by the user.

Q7. Ask the user for a temperature in Celsius (string input). Convert it to `float`, then calculate and print temperature in Fahrenheit.

Conversion formula: $FahrenheitTemp = (CelsiusTemp * (9/5)) + 32$

Q8. Take the radius (r) as user input and print the area.

Use the formula: $Area = \pi * r^2$ (value of π = 3.14)

Q9. Ask the user for: Principal (P), Rate (R), Time (T). Convert all to `float` and compute simple interest:

$$SI = (P * R * T)/100$$

Q10. Take a decimal number as input (like `45.78`) and output its:

- integer part - `45`
- fractional part - `.78`

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Assignment.py > ...

```python
name = input("Enter your name: ")
age = int(input("Enter your age: "))
print(f"Hello {name}, you are {age} years old!")
```

OUTPUT DEBUG CONSOLE TERMINAL PORTS

TERMINAL

```
PS C:\Users\USER\tryyyyyyyyyy> python -u "c:\Users\USER\tryyyyyyyyyy\Assignment.py"
Enter your name: Manoj Gharti
Enter your age: 20
Hello Manoj Gharti, you are 20 years old!
PS C:\Users\USER\tryyyyyyyyyy>
```

Assignment.py

Assignment.py > ...

```python
num1 = int(input("Enter First number: "))
num2 = int(input("Enter second number: "))
sum = num1 + num2
difference = num1 - num2
mul = num1 * num2
quotient = num1 / num2
print(f"The sum ,difference ,multiplication and quotient of {num1} and {num2} is {sum} ,{difference} ,{mul} and {quotient}")
```

OUTPUT DEBUG CONSOLE TERMINAL PORTS

TERMINAL

```
PS C:\Users\USER\tryyyyyyyyyy> python -u "c:\Users\USER\tryyyyyyyyyy\Assignment.py"
Enter First number: 45
Enter second number: 6
The sum ,difference ,multiplication and quotient of 45 and 6 is 51 ,39 ,270 and 7.5
PS C:\Users\USER\tryyyyyyyyyy>
```

Assignment.py > ...

```python
int_num1 = int(input("Enter first integer number: "))
int_num2 = int(input("Enter second integer number: "))
f_num = float(input("Enter float number: "))
avg = (float(int_num1) + float(int_num2) + f_num)/3
print(f"Avg = {avg}")
```

OUTPUT DEBUG CONSOLE TERMINAL PORTS

TERMINAL

```
PS C:\Users\USER\tryyyyyyyyyy> python -u "c:\Users\USER\tryyyyyyyyyy\Assignment.py"
Enter first integer number: 45
Enter second integer number: 24
Enter float number: 45.5
Avg = 38.166666666666664
PS C:\Users\USER\tryyyyyyyyyy>
```

Code

Code

Assignment.py

```python
string = input("Enter a number: ")
int_num = int(string)
float_num = float(string)
str_num = str(string)
print(f"Int_num = {int_num},{type(int_num)}\nFloat_num = {float_num},{type(float_num)}\nStr_num = {str_num},{type(str_num)}")
```

OUTPUT DEBUG CONSOLE TERMINAL PORTS

TERMINAL

```
PS C:\Users\USER\tryyyyyyyyyy> python -u "c:\Users\USER\tryyyyyyyyyy\Assignment.py"
Enter a number: 45
Int_num = 45,<class 'int'>
Float_num = 45.0,<class 'float'>
Str_num = 45,<class 'str'>
PS C:\Users\USER\tryyyyyyyyyy>
```

Code

Code

Assignment.py

Assignment.py > ...

```python
x = 10 + 3 * 2 ** 2 #According to Operator Precedence rule (** -> * -> +)so output is 22 let's print
print(x)
```

OUTPUT DEBUG CONSOLE TERMINAL PORTS

TERMINAL

```
PS C:\Users\USER\tryyyyyyyyyy> python -u "c:\Users\USER\tryyyyyyyyyy\Assignment.py"
22
PS C:\Users\USER\tryyyyyyyyyy>
```

Code

Code

Assignment.py

Assignment.py > ...

```python
num1 = int(input("Enter first number: "))
num2 = int(input("Enter second number: "))
print(f"Before Swaping:\nnum1 = {num1} and num2 = {num2}")
num1 = num1 + num2
num2 = num1 - num2
num1 = num1 - num2
print(f"After Swaping:\nnum1 = {num1} and num2 = {num2}")
```

OUTPUT DEBUG CONSOLE TERMINAL PORTS

TERMINAL

```
PS C:\Users\USER\tryyyyyyyyyy> python -u "c:\Users\USER\tryyyyyyyyyy\Assignment.py"
Enter first number: 45
Enter second number: 34
Before Swaping:
num1 = 45 and num2 = 34
After Swaping:
num1 = 34 and num2 = 45
PS C:\Users\USER\tryyyyyyyyyy>
```

Code

Code

Assignment.py

Assignment.py > ...

```python
c_tem = float(input("Enter tempreture in celsius: "))
f_tem = (c_tem * (9 / 5)) + 32
print(f"C_tem = {c_tem}\nF_tem = {f_tem}")
```

OUTPUT DEBUG CONSOLE TERMINAL PORTS

TERMINAL

```
PS C:\Users\USER\tryyyyyyyyyy> python -u "c:\Users\USER\tryyyyyyyyyy\Assignment.py"
Enter tempreture in celsius: -40
C_tem = -40.0
F_tem = -40.0
PS C:\Users\USER\tryyyyyyyyyy> python -u "c:\Users\USER\tryyyyyyyyyy\Assignment.py"
Enter tempreture in celsius: -39.9999999
C_tem = -39.9999999
F_tem = -39.99999982
PS C:\Users\USER\tryyyyyyyyyy> python -u "c:\Users\USER\tryyyyyyyyyy\Assignment.py"
Enter tempreture in celsius: -38
C_tem = -38.0
F_tem = -36.400000000000006
PS C:\Users\USER\tryyyyyyyyyy>
```

Code

Code

Assignment.py

Assignment.py > ...

```python
radius = int(input("Enter radius of circle: "))
pi = 22/7
print(f"Area of Circle is {pi*radius**2}")
```

OUTPUT DEBUG CONSOLE TERMINAL PORTS

TERMINAL

```
PS C:\Users\USER\tryyyyyyyyyy> python -u "c:\Users\USER\tryyyyyyyyyy\Assignment.py"
Enter radius of circle: 7
Area of Circle is 154.0
PS C:\Users\USER\tryyyyyyyyyy>
```

Code

Code

Assignment.py

Assignment.py > ...

```python
P = float(input("Enter principal: "))
T = float(input("Enter time: "))
R =float(input("Enter rate: "))
S_I = (P * T * R) / 100
print(f"S.I = {S_I}")
```

OUTPUT DEBUG CONSOLE TERMINAL PORTS

TERMINAL

```
PS C:\Users\USER\tryyyyyyyyyy> python -u "c:\Users\USER\tryyyyyyyyyy\Assignment.py"
Enter principal: 10000000
Enter time: 23
Enter rate: 3
S.I = 6900000.0
PS C:\Users\USER\tryyyyyyyyyy>
```

Code

Code

Assignment.py

Assignment.py > ...

```python
dec_num = 45.78
int_part = int(dec_num)
frac_part = dec_num - int_part
print(f"Integer Part: {int_part}\nFractional Part: {round(frac_part,2)}")
```

OUTPUT DEBUG CONSOLE TERMINAL PORTS

TERMINAL

```
PS C:\Users\USER\tryyyyyyyyyy> python -u "c:\Users\USER\tryyyyyyyyyy\Assignment.py"
Integer Part: 45
Fractional Part: 0.78
PS C:\Users\USER\tryyyyyyyyyy>
```

Code

Code